Supplementary Information

Carrier asymmetry and thermal-model dependence in neural transport decomposition

Model names correspond to the physical hypotheses defined in the main text. All coverage values are nominal 95% pointwise coverage conditional on scored cases. Native populations can differ; common-case comparisons use explicit intersections.

Supplementary Table 1 | Independent same-model coverage

Model	Scored / generated	Admitted	Lattice coverage percent	Bipolar coverage percent
Independent + flexible	5986/6000	5842	91.7	92.0
Independent + rising	5989/6000	5882	93.3	91.7
Independent bands	5998/6000	5856	94.7	91.8
Flexible lattice	5992/6000	5847	90.9	92.1
Rising channel	5993/6000	5887	93.9	92.1

Supplementary Table 2 | Native narrow-gap upturn evaluation

Model	Scored	Admitted	Lattice coverage percent	Bipolar coverage percent	Slope subset n	Median slope error
Independent + flexible	999	998	97.2	92.9	343	-0.019
Independent + rising	656	634	36.2	91.6	182	-0.702
Independent bands	599	577	40.7	82.6	154	-0.693
Flexible lattice	999	985	95.4	88.4	359	-0.058
Rising channel	619	571	37.3	85.3	174	-0.660

Supplementary Table 3 | Experimental slope comparisons

Model	Native paired n	Native C minus B	All-model-common C minus B	Paired change from Reference
Reference	4312	-0.165 [-0.184, -0.150]	-0.107 [-0.122, -0.092]	0.000 [0.000, 0.000]
Independent + flexible	4187	-0.452 [-0.499, -0.407]	-0.593 [-0.702, -0.481]	-0.425 [-0.540, -0.354]
Independent + rising	4507	-0.302 [-0.322, -0.287]	-0.227 [-0.242, -0.208]	-0.094 [-0.102, -0.087]
Independent bands	5368	-0.209 [-0.222, -0.194]	-0.145 [-0.165, -0.132]	-0.018 [-0.024, -0.014]
Flexible lattice	4225	-0.577 [-0.683, -0.504]	-0.373 [-0.456, -0.308]	-0.231 [-0.284, -0.188]
Rising channel	3114	-0.283 [-0.308, -0.254]	-0.181 [-0.199, -0.163]	-0.065 [-0.070, -0.059]

Supplementary Table 4 | Common-case upturn accuracy and interval width

Model	Lattice coverage percent	Median interval width	Median slope error	Median absolute slope error
Reference	41.7	0.699	-0.541	0.541
Independent + flexible	97.2	9.983	-0.241	0.661
Independent + rising	42.5	0.898	-0.600	0.600
Independent bands	45.4	0.972	-0.594	0.594
Flexible lattice	95.2	6.446	-0.606	0.822
Rising channel	41.3	0.751	-0.562	0.562

Supplementary Table 5 | Common-case mobility-prefactor stress

Model	Mobility ratio	Common scored n	Lattice coverage percent	Bipolar coverage percent
Reference	0.1	790	91.8	75.5
Independent + flexible	0.1	790	98.3	82.7
Independent + rising	0.1	790	89.9	91.2
Independent bands	0.1	790	95.6	89.4
Flexible lattice	0.1	790	92.8	72.1
Rising channel	0.1	790	86.1	77.0
Reference	10	648	91.4	74.9
Independent + flexible	10	648	98.4	86.0
Independent + rising	10	648	89.9	91.7
Independent bands	10	648	96.5	90.5
Flexible lattice	10	648	92.8	70.2
Rising channel	10	648	86.6	76.9

Supplementary Table 6 | Transfer between thermal truth models

Inference model	Truth model	Scored / generated	Lattice coverage percent	Bipolar coverage percent
Reference	Independent + rising	662/1000	58.1	69.3
Reference	Independent + flexible	164/1000	21.8	65.1
Independent + flexible	Independent + rising	999/1000	74.5	91.6
Independent + flexible	Independent + flexible	996/1000	90.2	95.5
Independent + rising	Independent + rising	999/1000	92.6	94.9
Independent + rising	Independent + flexible	268/1000	25.8	85.9

Inference model	Truth model	Scored / generated	Lattice coverage percent	Bipolar coverage percent
Independent bands	Independent + rising	867/1000	74.8	84.6
Independent bands	Independent + flexible	312/1000	25.8	80.8
Flexible lattice	Independent + rising	781/1000	62.6	70.8
Flexible lattice	Independent + flexible	741/1000	82.4	76.2
Rising channel	Independent + rising	743/1000	83.7	75.2
Rising channel	Independent + flexible	145/1000	23.9	68.7

Supplementary Table 7 | Effective-scattering ambiguity

Model	Admitted n	Ambiguous majority/shared exponent	Ambiguous minority exponent
Independent + flexible	10740	5575	9445
Independent + rising	8210	2362	6082
Independent bands	9604	2621	7891
Flexible lattice	7736	1975	shared parameter
Rising channel	5623	792	shared parameter

Supplementary Table 8 | Neural training completion

Epochs are one-based. The selected checkpoint is the best qualifying validation checkpoint. These are single-seed runs, not replicated convergence assessments.

Variant	Completed epochs	Selected epoch	Stop reason	Seed
Independent bands	178	128	patience	6101
Rising channel	618	568	patience	6111
Flexible lattice	900	899	epoch_cap	6121
Independent + rising	264	214	patience	6131
Independent + flexible	214	164	patience	6141

Full per-curve inference outputs, comparison masks, nested matched-draw summaries, subgroup summaries and software provenance are supplied in the repository. The prior-support adaptation and inherited development-benchmark status are described in Methods. No simulation-based calibration ranks or external validation measurements are supplied.